

LAB NO#01

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(CT-25169)

Question no#01:

//1. Write a C++ program to copy data of a 2D array in a 1D array using Column Major Order.

```
#include<iostream>

using namespace std;

int main(){

    int r,c;

    cout<<"enter rows:";

    cin>>r;

    cout<<"enter columns:";

    cin>>c;


    int array2D[r][c];

    cout<<"enter elements of 2D array:"<<endl;

    for(int i=0;i<r;i++){

        for(int j=0;j<c;j++){

            cin>>array2D[i][j];

        }

    }
```

```

    }

    int array1D[r*c];

    int k=0;

    for(int i=0;i<c;i++){
    for(int j=0;j<r;j++){
    array1D[k] = array2D[j][i];
    k++;
    }
};

    cout<<"elements of 1D array become";

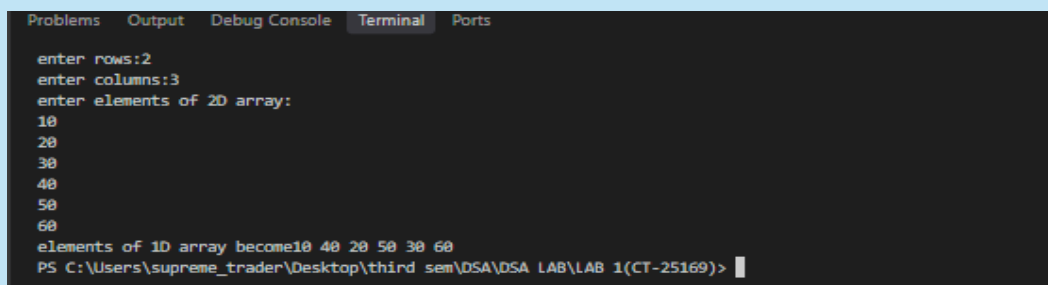
    for(int i=0;i<r*c;i++){

        cout<<array1D[i]<<" ";

    }

}

```



The screenshot shows a terminal window with the following input and output:

```

enter rows:2
enter columns:3
enter elements of 2D array:
10
20
30
40
50
60
elements of 1D array become10 40 20 50 30 60
PS C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CT-25169)>

```

Question no#02:

//Write a program to calculate the GPA of students of all subjects of a single semester .

//Assume all the courses have the same credit hour (let's assume 3 credit hours).

```
#include<iostream>

using namespace std;

int main(){

    string name[]={"Ali","Hiba","Asma","Zain","Faisal"};

    //for "--" we take -1.0

    float marks[5][5]={

        {3.66, 3.33, 4.0, 3.0, 2.66},

        {3.33, 3.0, 3.66, 3.0, -1.0},

        {4.0, 3.66, 2.66, -1.0, -1.0},

        {2.66, 2.33, 4.0, -1.0, -1.0},

        {3.33, 3.66, 4.0, 3.0, 3.33}

    };

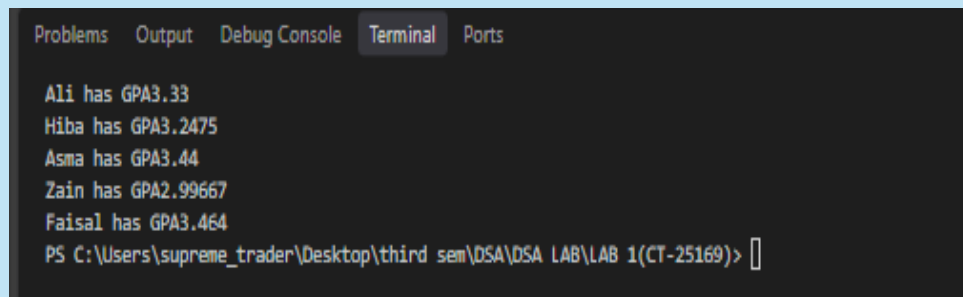
    for(int i=0;i<5;i++){

        float sum=0.0;

        int no_of_valid_sub=0;

        for(int j=0;j<5;j++){
```

```
        if(marks[i][j]!=-1.0){  
            sum=sum+marks[i][j];  
            no_of_valid_sub++;  
        }  
    }  
  
    float gpa=sum/no_of_valid_sub;  
    cout<<name[i]<<" has GPA"<<gpa<<endl;  
  
}  
  
}
```



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal' (which is selected), and 'Ports'. Below the tabs, the output of the program is displayed as five lines of text, each showing a name followed by 'has GPA' and a numerical value. The names are Ali, Hiba, Asma, Zain, and Faisal. The GPA values are 3.33, 3.2475, 3.44, 2.99667, and 3.464 respectively. At the bottom of the terminal, there is a command prompt showing the current directory as 'C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CT-25169)' followed by a greater-than sign and a cursor.

```
Problems  Output  Debug Console  Terminal  Ports  
  
Ali has GPA3.33  
Hiba has GPA3.2475  
Asma has GPA3.44  
Zain has GPA2.99667  
Faisal has GPA3.464  
PS C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CT-25169)> 
```

Question no#03:

/*The median is the middle value in an ordered integer list.

If the size of the list is even, there is no middle value, and the median is the mean of the two middle values. For example, for arr = [2,3,4], the median is 3.

For example, for arr = [2,3], the median is $(2 + 3) / 2 = 2.5$. Implement the MedianFinder class: - MedianFinder() initializes the MedianFinder object. - void addNum(int num) adds

the integer num from the data stream to the data structure. - double findMedian() returns the median of all elements so far. Answers within 10⁻⁵ of the actual answer will be accepted.

**Example 1: Input: ["MedianFinder", "addNum", "addNum", "findMedian", "addNum", "findMedian"]
Output: [null, [1], [2], [1.5], [3], [2.0]]**

Explanation
`MedianFinder medianFinder = new MedianFinder();
medianFinder.addNum(1); // arr = [1]
medianFinder.addNum(2); // arr = [1, 2]
medianFinder.findMedian(); // return 1.5 (i.e., $(1 + 2) / 2$)
medianFinder.addNum(3); // arr[1, 2, 3]
medianFinder.findMedian(); // return 2.0`
Constraints: -10⁵ ≤ num ≤ 10⁵ There will be at least one element in

the data structure before calling findMedian.

At most 5 * 10⁴ calls will be made to addNum and findMedian. */

```
#include <iostream>
```

```
#include<vector>
```

```
#include <algorithm>
```

```
using namespace std;
```

```
class MedianFinder{
    private:
    vector<int>numb;
    public:
    //constructor
    MedianFinder(){}

    void addnum(int num){
        numb.push_back(num);//it adds data elements
    sort(numb.begin(),numb.end());
    }

    double findMedian(){
        int n=numb.size();

        if(n%2!=0){//for odd
return numb[n/2];
        }
        else{//for even
            return(numb[n/2-1]+numb[n/2])/2.0;
        }
    }
};

int main(){
    MedianFinder med;
```

```

    med.addnum(1);

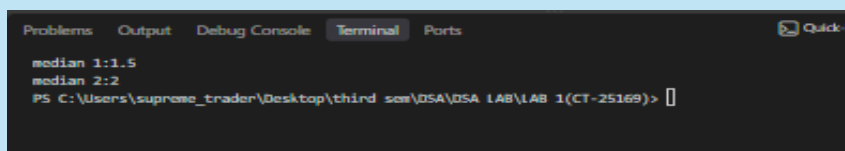
    med.addnum(2);

    cout<<"median 1:"<<med.findMedian()<<endl;
med.addnum(3);

    cout<<"median 2:"<<med.findMedian()<<endl;

}

```



```

median 1:1.5
median 2:2
PS C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CY-25169)>

```

Question no#04:

Given an array of integers nums which is sorted in ascending order, and an integer target, write a function to search target in nums. If target exists, then return its index. Otherwise, return -1. You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1: Input: nums = [-1,0,3,5,9,12], target = 9, Output: 4 Explanation: 9 exists in nums and its index is 4

Example 2: Input: nums = [-1,0,3,5,9,12], target = 2, Output: -1 Explanation: 2 does not exist in nums so return -1

Constraints: $1 \leq \text{nums.length} \leq 10^4$ $-10^4 < \text{nums}[i], \text{target} < 10^4$ All the integers in nums are unique.

/*Given an array of integers nums which is sorted in

ascending order, and an integer target, write a function to search target in nums.

If target exists, then return its index. Otherwise, return -1. You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1: Input: nums = [-1,0,3,5,9,12], target = 9, Output: 4

Explanation: 9 exists in nums and its index is 4

Example 2: Input: nums = [-1,0,3,5,9,12], target = 2, Output: -1

Explanation: 2 does not exist in nums so return -1

Constraints: $1 \leq \text{nums.length} \leq 10^4$, $\text{nums}[i] < \text{nums}[i+1]$, target $\in [-10^4, 10^4]$ All the integers in nums are unique.*

```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
int search(vector<int>&nums,int target){
```

```
    int l=0;
```

```
    int r=nums.size()-1;
```

```
    while(l<=r){
```

```
        int mid=l+(r-l)/2;
```

```
    if(nums[mid]==target){  
        return mid;  
    }  
    else if(nums[mid]<target){  
        l=mid+1;  
    }  
    else{  
        r=mid-1;  
    }  
}  
return -1;  
  
}
```

```
int main(){
```

```
    vector<int> nums = {-1, 0, 3, 5, 9, 12};
```

```
    int target1= 9;
```

```
    int result = search(nums, target1);
```

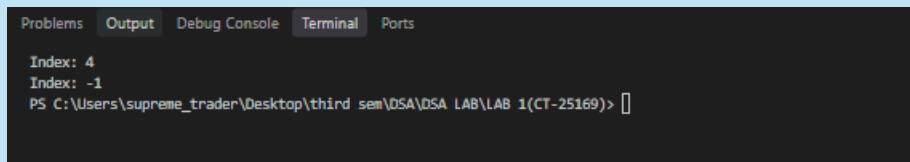
```
    cout << "Index: " << result << endl; // Output: 4
```

```
    int target2 = 2;
```

```
    int result2 = search(nums, target2);
```

```
    cout << "Index: " << result2 << endl; // Output: -1
```

```
    return 0;
}
```



```
Problems Output Debug Console Terminal Ports
Index: 4
Index: -1
PS C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CT-25169)> []
```

Question no#05:

nums is sorted in ascending order. You are given an $m \times n$ integer matrix with the following two properties: Each row is sorted in non-decreasing order. The first integer of each row is greater than the last integer of the previous row. Given an integer target, return true if target is in matrix or false otherwise. You must write a solution in $O(\log(m * n))$ time complexity. Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3, Output: true Constraints: $m == \text{matrix.length}$, $n == \text{matrix}[i].\text{length}$, $1 \leq m$, $n \leq 100$, $-104 \leq \text{matrix}[i][j]$, target ≤ 104

```
/*nums is sorted in ascending order. You are given an m x n integer matrix with
the following two properties: Each row is sorted in non-decreasing order. The
first integer of each row is greater than the last integer of the previous row.
Given an integer target, return true if target is in matrix or false otherwise. You
must write a solution in O(log(m * n)) time complexity. Input: matrix =
[[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3, Output: true Constraints: m ==
matrix.length, n == matrix[i].length, 1 <= m, n <= 100, -104 <= matrix[i][j],
target <= 104 */
```

```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
bool searchMatrix(vector<vector<int>>&matrix,int target){
```

```
    int m=matrix.size();
```

```
    int n=matrix[0].size();
```

```
    int l = 0;
```

```
    int r = (m * n) - 1;
```

```
    while (l <= r) {
```

```
        int mid = l + (r - l) / 2;
```

```
        int row = mid / n;
```

```
        int col = mid % n;
```

```
        if (matrix[row][col] == target) {
```

```
            return true;
```

```
        }
```

```
        else if (matrix[row][col] < target) {
```

```
            l = mid + 1;
```

```
        }
```

```
        else {
```

```
            r = mid - 1;
```

```
    }  
}  
  
    return false;  
}  
  
int main() {  
    vector<vector<int>> matrix = {  
        {1, 3, 5, 7},  
        {10, 11, 16, 20},  
        {23, 30, 34, 60}  
    };  
  
    int target1 = 3;  
  
    cout << (searchMatrix(matrix, target1) ? "true" : "false") << endl; // Output:  
true  
  
    int target2 = 13;  
  
    cout << (searchMatrix(matrix, target2) ? "true" : "false") << endl; // Output:  
false  
  
    return 0;  
}
```

Problems Output Debug Console **Terminal** Ports

true

false

PS C:\Users\supreme_trader\Desktop\third sem\DSA\DSA LAB\LAB 1(CT-25169)> |